
Fwd: the paper you helped me with

9 件のメッセージ

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月21日 5:59

To: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Hi,
here are questions that were sent by Bob Cousins. I think the answer to all of them is that we do not know. But before answering maybe we need to think and discuss them.

Graciela

----- Forwarded message -----

From: **Robert Cousins** <cousins@physics.ucla.edu>

Date: Thu, Aug 20, 2026 at 1:20 PM

Subject: Re: the paper you helped me with

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Hi Graciela,

I managed to read a few pages at the beginning and a few pages at the end, I will not be able to try to understand it in any detail. So let me just say a few things about unfolding, which I expect we have discussed.

Any unfolding technique can be characterised by the bias of the estimates and the variance of the estimates. A famous (trivial) theorem is that the mean squared error (MSE) with respect to the true value is equal to the sum of the variance (with respect to the sample mean) and the square of the bias. The whole game in unfolding is to optimize the MSE by trading off bias and variance, The idea behind all regularization, including Tikhonov, is to introduce a small bias in order to reduce the variance by a lot, and hence win on MSE.

So one question I have is: do you understand the bias-variance tradeoff in your method? Are you getting zero bias by construction, and hence a large variance?

Another big property in unfolding is correlations among the unfolded quantities. They exist in all defensible unfolding techniques. Do you understand correlations in your application?

Best,
Bob

On Mon, Aug 17, 2026 at 12:52 PM Graciela Gelmini <gelmini@physics.ucla.edu> wrote:

Thanks Bob for your prompt answer. Enjoy your holidays!!!

Do not worry about reading the paper urgently!!!!

We do not understand why the bands seem to increase little with background level, or if with Tikhonov-regularized unfolding they also change equally little (since in the paper using this method for the low energy nu background this comparison was not done).

We do not even know exactly how to find out the reason for this feature.

Thanks again! Warm greeting to Lynne!

Graciela

----- Forwarded message -----

From: **Robert Cousins** <cousins@physics.ucla.edu>

Date: Mon, Aug 17, 2026 at 3:01 AM

Subject: Re: the paper you helped me with

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Hi Graciela,

Lynne and I are in the Swiss Alps with Eleri's family. I will try to look at this when I get a chance, but it will be a few days,

Before looking at it, I would say that the question is not whether insensitivity to the background level is a "good thing" (where good means convenient). The question is whether it is a "sensible thing". I would say that in many cases, it means a bug. One special case where it is correct and well understood is a Bayesian analysis where there are no observed events at all. In that case, the upper limit is independent of the expected mean background. In your case, it is important to understand why you are insensitive -- I am sure you realise this.

Best,
Bob

Best,
Bob

On Sun, Aug 16, 2026 at 1:32 PM Graciela Gelmini <gelmini@physics.ucla.edu> wrote:

Dear Bob,

I hope you and your family are all well. Have not heard from you in a while.

Here is finally the paper you help me with. We sent it to the arXiv and will appear in the next few days.

We call the method Finite Dirac Sum unfolding, It is different from Tikhonov-regularized unfolding, without all the arbitrary choices of smoothing inherent in it. It does not require smoothing.

The main problem of the method as it is is that the fit and two-sided bands we found are for the integrated neutrino flux and not for the differential flux. We did not do the work to find fits and bands in the differential flux, because the whole thing was a lot of work. so at some point one declares the paper done.

There is an issue we discussed with Danny Marfatia and, without any urgency, I would like to know your opinion about it. Our pointwise confidence bands are relatively insensitive to the level of background (we tested it keeping all other characteristics the same and just changing the background). I thought this is a drawback (which also Tikhonov-regularized unfolding may have, if the smoothness required is strong- but we did not find this explicitly tested). Danny Marfatia said that it could instead be a good thing, namely that knowing well your background is not important. Finally we settled on just mentioning it in the paper as a feature (not a drawback as I had written initially).

What do you think, a relative insensitivity to the level of background is a problem for the method, or not?

I want to thank you a lot again because without your help about the Monte Carlo procedure we could not have done this paper.

Best,
Graciela

To: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Hello, this is what Bob said of how one determines the bias of the method.
I do not understand his proposal at the moment. Which answer one has to make a histogram of? I understand what is the variance of a histogram, but what is the bias?

I do not think we want to do this now in any event....

Graciela

----- Forwarded message -----

From: **Robert Cousins** <cousins@physics.ucla.edu>
Date: Fri, Aug 21, 2026 at 1:58 AM
Subject: Re: the paper you helped me with
To: Graciela Gelmini <gelmini@physics.ucla.edu>

Hi Graciela,

The way to determine the bias is with Monte Carlo simulation, Create an ensemble of pseudo-experiments, run your algorithm on each pseudo-experiment, and histogram the distribution of each answer. The bias and variance can be read off the histograms.

B-BC

On Thu, Aug 20, 2026 at 2:38 PM Graciela Gelmini <gelmini@physics.ucla.edu> wrote:

Thanks a lot Bo!.

These are extremely good questions, which I do not know the answers to. I have shared them with my collaborators.

How can we determine the bias of our unfolding method?

We do not have any regularization function, so I do not think we can modify the bias, whatever it is in our procedure. The "regularization" if it can be described as one, is done through the monotonicity condition on the integrated flux, i.e. requiring it to be non-increasing with increasing energy (a physical requirement).

Best,

Graciela

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Danny Marfatia <dmarf8@hawaii.edu>

2026年8月23日 14:13

To: Graciela Gelmini <gelmini@physics.ucla.edu>, cousins@physics.ucla.edu

Cc: Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Hi Bob, I am going to jump in as a coauthor on the paper you are discussing with Graciela. The bias we discuss in the paper is specific to the smoothing function and regularization parameter of Tikhonov unfolding; see Eqs. (11.76) and (11.83) of Cowan's book (attached). Note that in my paper with Graciela, there is no regularization at all, so is it fair to say that the bias you have been discussing is of a different nature, related to how the datasets are simulated?

Thanks, Danny.



Muping Chen <mpchen1864@gmail.com>

2026年8月23日 15:13

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Danny Marfatia <dmarf8@hawaii.edu>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Hi Graciela,

Please take a look at the attached screenshots.

It seems bias can mean two different things.

What we have so far is scenario B.

But I think Cousins asked for scenario A.

However, I am not sure which histogram he is referring to, since we have many procedures involving making histograms.

Maybe he can clarify it a little bit?

Best Regards,

Muping Chen

Graciela Gelmini <gelmini@physics.ucla.edu> 于2026年8月22日周六 05:05写道：

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添付ファイル 2 件



Screenshot 2026-08-23 144948.png

118K



Screenshot 2026-08-23 144940.png

154K

Robert Cousins <cousins@physics.ucla.edu>

2026年8月23日 19:14

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro

<k.yasuda5160@gmail.com>

Dear Danny, all,

Thanks for the email. In my experience, if stats questions are not resolved after a couple emails, then there is probably some sort of misunderstanding, and a Zoom to chat about it may be more useful. That may be the case here. So I give a quick reply and if that is not clear, then maybe we can talk next week.

As I understand it, your paper is making point estimates of something. The bias I am talking about is Eq. 5.4 of Glen's book in the chapter about point estimation. Eq. 5.5 is what I was referring to when I told Graciela on August 20, "A famous (trivial) theorem is that the mean squared error (MSE) with respect to the true value is equal to the sum of the variance (with respect to the sample mean) and the square of the bias."

This is all very general, long before Glen talks about unfolding. Bias can come from lots of places, not just regularization in unfolding. In general, maximum likelihood estimates (MLEs) are biased, for example. As a famous simple example, if you have n samples from a Gaussian with unknown mean and unknown variance, the MLE is biased: it has $1/n$ instead of the $1/(n-1)$ that Glen puts in Eq. 5.9. See Eqs. 6.12 and 6.13 and discussion there. And as I mentioned, the bias changes as you change metrics. Note that the square root of an unbiased estimate of the variance is a biased estimate of the RMS deviation. So bias can be confusing (and Bayesians claim not to care about it). Using the median rather than the mean has the advantage that it is independent of the metric, but it is not as convenient. Fortunately, all these issues usually go away as $1/n$.

So, my suggestion was to histogram your point estimates (θ -hat in the notation of Eqn. 5.4) and difference from true θ , so that you can see by eye if it is biased and what the variance is. In general, one always wants to know the bias and variance of point estimates. In your case, especially, since your papers refers to what you do as "unfolding" (as opposed to "fitting", for example), and given that the bias-variance tradeoff is famous in unfolding, I think that readers will be especially interested to know the bias and variance of your point estimates.

Probably it is now clear that my answer is "No" to your question, "so is it fair to say that the bias you have been discussing is of a different nature, related to how the datasets are simulated?"

If you would like to Zoom, I am usually available from early morning in L.A. until noon or so.

Best,
Bob

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Danny Marfatia <dmarf8@hawaii.edu>

2026年8月23日 19:31

To: Robert Cousins <cousins@physics.ucla.edu>

Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Thanks for the response, although the bias in Tikhonov unfolding arising from regularization does still seem to be different to me. I am at CERN currently, so if you are around next week, there's no need to Zoom!

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Robert Cousins <cousins@physics.ucla.edu>

2026年8月23日 19:47

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Yes, I am happy to have coffee at CERN -- I can be in touch next week regarding possibilities. I am not clear why you mention Tikhonov unfolding now. I mentioned it in passing while mentioning the bias-variance tradeoff in unfolding, but nothing I said about your method was intended to have anything to do with Tikhonov unfolding, just properties of your method. -BC

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Danny Marfatia <dmarf8@hawaii.edu>

2026年8月23日 19:56

To: Robert Cousins <cousins@physics.ucla.edu>

Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Ah, my original question was about bias in Tikhonov unfolding (which is mentioned in our paper) versus the bias being discussed by Graciela and you in your email exchange. My question was whether these two are different because the arbitrariness in Tikhonov regularization is inherently biased.

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Robert Cousins <cousins@physics.ucla.edu>

2026年8月23日 20:20

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

The answer to that may lie in understanding the properties of your method. It may be partly semantics. Experts consider binning of histograms to be a form of regularization. The architecture of a neural net can introduce regularization, even with no obvious tunable "knob". I do not if your method implicitly regularizes. (Sorry for making general statements, as I am not able to find time to read your paper carefully, with various things going on.)

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